

Some Basic Concepts Of Chemistry

Question1

Dalton's Atomic theory could not explain which of the following?

NEET 2025

Options:

- A. Law of multiple proportion
- B. Law of gaseous volume
- C. Law of conservation of mass
- D. Law of constant proportion

Answer: B

Solution:

Dalton's theory could explain the laws of chemical combination. However, it could not explain the laws of gaseous volumes.

Question2

Among the following, choose the ones with equal number of atoms.

- A. 212 g of $\text{Na}_2\text{CO}_3(\text{s})$ [molar mass = 106 g]
- B. 248 g of $\text{Na}_2\text{O}(\text{s})$ [molar mass = 62 g]
- C. 240 g of $\text{NaOH}(\text{s})$ [molar mass = 40 g]

D. 12 g of H_2 (g) [molar mass = 2 g]

E. 220 g of CO_2 (g) [molar mass = 44 g]

Choose the correct answer from the options given below:

NEET 2025

Options:

A. B, C, and D only

B. B, D, and E only

C. A, B, and C only

D. A, B, and D only

Answer: D

Solution:

$$\text{Number of atoms} = \frac{\text{given mass}}{\text{molar mass}} \times \text{atomicity} \times N_A$$

A. $\frac{212}{106} \times 6 \times N_A = 12 N_A$

B. $\frac{248}{62} \times 3 \times N_A = 12 N_A$

C. $\frac{240}{40} \times 3 \times N_A = 18 N_A$

D. $\frac{12}{2} \times N_A \times 2 = 12 N_A$

E. $\frac{220}{44} \times N_A \times 3 = 15 N_A$

A, B and D have same number of atoms

Question3

The amount of glucose required to prepare 250 mL of M/20 aqueous solution is :

(Molar mass of glucose : 180g mol^{-1})

[NEET 2024 Re]

Options:

A.

2.25 g

B.

4.5 g

C.

0.44 g

D.

1.125 g

Answer: A

Solution:

$$\text{Molarity, } M = \frac{w_2 \times 1000}{M_2 \times (V)}$$

w_2 = Amount of glucose

$$\text{Given molarity} = \frac{M}{20}$$

$$\frac{1}{20} = \frac{w_2 \times 1000}{180 \times 250}$$

$$w_2 = \frac{180 \times 250}{20 \times 1000}$$

$$= 2.25\text{g}$$

Question4

1.0 g of H_2 has same number of molecules as in:

[NEET 2024 Re]

Options:

A.

14 g of N_2

B.

18 g of H_2O

C.

16 g of CO

D.

28 g of N_2

Answer: A

Solution:

$$\text{Number of moles of } H_2 = \frac{1}{2} = 0.5$$

$$\text{Number of molecules of } H_2 = 0.5N_A$$

$$(1) \text{ Number of moles of } N_2 = \frac{14}{28} = 0.5$$

$$\text{Number of molecules of } N_2 = 0.5N_A$$

$$(2) \text{ Number of moles of } H_2O = \frac{18}{18} = 1$$

$$\text{Number of molecules of } H_2O = 1 \times N_A = N_A$$

$$(3) \text{ Number of moles of CO} = \frac{16}{28} = \frac{4}{7}$$

$$\text{Number of molecules of CO} = \frac{4}{7}N_A$$

$$(4) \text{ Number of moles of } N_2 = \frac{28}{28} = 1$$

$$\text{Number of molecules of } N_2 = 1 \times N_A = N_A$$

Question5

On complete combustion, 0.3 g of an organic compound gave 0.2 g of CO_2 and 0.1 g of H_2O . The percentage composition of carbon and hydrogen in the compound, respectively is:

[NEET 2024 Re]

Options:

A.

4.07% and 15.02%

B.

18.18% and 3.70%

C.

15.02% and 4.07%

D.

3.70% and 18.18%

Answer: B

Solution:

$$\text{Percentage of carbon} = \frac{12 \times m_2 \times 100}{44 \times m}$$

$$m = \text{mass of organic compound} = 0.3\text{g}$$

$$m_2 = \text{mass of carbon dioxide} = 0.2\text{g}$$

$$\therefore \%C = \frac{12 \times 0.2 \times 100}{44 \times 0.3} = 18.18\%$$

$$\text{Percentage of hydrogen} = \frac{2 \times m_1 \times 100}{18 \times m}$$

$$m_1 = \text{mass of water} = 0.1\text{g}$$

$$\therefore \%H = \frac{2 \times 0.1 \times 100}{18 \times 0.3} = 3.70\%$$

Question6

1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to

[NEET 2024]

Options:

A.

750 mg

B.

250 mg

C.

Zero mg

D.

200 mg

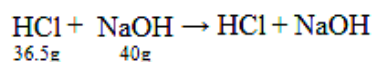
Answer: B

Solution:

$$M = \frac{W \times 1000}{M_2 \times V(\text{ in mL})}$$

$$W = \frac{M \times M_2 \times V(\text{ in mL})}{1000} = \frac{0.75 \times 36.5 \times 25}{1000}$$

$$= 0.684\text{g (Mass of HCl)}$$



36.5g HCl reacts with NaOH = 40g

$$0.684\text{g HCl reacts with NaOH} = \frac{40}{36.5} \times 0.684 \approx 0.750\text{g}$$

$$\text{Amount of NaOH left} = 1\text{g} - 0.750\text{g} = 0.250\text{g} = 250\text{mg}$$

Question7

The highest number of helium atoms is in

[NEET 2024]

Options:

A.

4 mol of helium

B.

4u of helium

C.

4g of helium

D.

2.271098L of helium at STP

Answer: A

Solution:

$$(1) 4 \text{ mol of He} = 4 N_A \text{ He atoms}$$

$$(2) 4 \text{ u of He} = \frac{4 \text{ u}}{4 \text{ u}} = 1 \text{ He atom}$$

$$(3) 4 \text{ g of Helium} = \frac{4 \text{ g}}{4 \text{ g}} \text{ mole} = 1 \text{ mole} = N_A \text{ He atom}$$

$$(4) 2.2710982 \text{ L of He at STP} = \frac{2.271}{22.710982} \text{ mole}$$

$$= 0.1 \text{ mole}$$

$$= 0.1 N_A \text{ He atom}$$

Question8

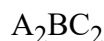
A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of A = 64 ; B = 40 ; C = 32u)

[NEET 2024]

Options:

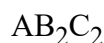
A.



B.



C.



D.



Answer: B

Solution:

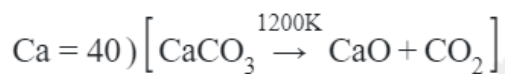
Element	Masspercentage %	No. of moles	No. of moles/Smallest number	Simplest whole number
A	32%	$\frac{32}{64} = \frac{1}{2}$	$\frac{1}{2} \times 2$	= 1
B	20%	$\frac{20}{40} = \frac{1}{2}$	$\frac{1}{2} \times 2$	= 1
C	48%	$\frac{48}{32} = \frac{3}{2}$	$\frac{3}{2} \times 2$	= 3

So, empirical formula of X = $\begin{matrix} A & : & B & : & C \\ 1 & : & 1 & : & 3 \end{matrix}$

$\therefore$ The correct empirical formula of compound X is ABC_3

Question9

The right option for the mass of CO_2 produced by heating 20g of 20% pure limestone is (Atomic mass of



[NEET 2023]

Options:

A.

1.76g

B.

2.64g

C.

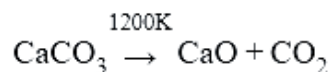
1.32g

D.

1.12g

Answer: A

Solution:



From $100\text{gCaCO}_3 \rightarrow 44\text{gCO}_2$ produced

As CaCO_3 is 20% pure

$$\text{So, mass of pure CaCO}_3 = 20 \times \frac{20}{100} = 4\text{g}$$

So, $100\text{gCaCO}_3 \rightarrow 44\text{gCO}_2$

$$4\text{gCaCO}_3 \rightarrow \frac{44}{100} \times 4\text{gCO}_2$$

$$= 1.76\text{gCO}_2$$

Question10

The density of 1M solution of a compound ' X ' is 1.25gmL^{-1} . The correct option for the molality of solution is (Molar mass of compound X = 85g) :

[NEET 2023 mpr]

Options:

A.

0.705m

B.

1.208m

C.

1.165m

D.

0.858m

Answer: D

Solution:

$$m = \frac{1000 \times M}{1000 \times d - MM_w}$$

$$m = \frac{1000 \times 1}{1000 \times 1.25 - 1 \times 85}$$

$$m = \frac{1000}{1165} = 0.858$$

Question11

What mass of 95% pure CaCO_3 will be required to neutralise 50mL of 0.5M HCl solution according to the following reaction?



[Calculate upto second place of decimal point]

[NEET-2022]

Options:

A. 1.25g

B. 1.32g

C. 3.65g

D. 9.50g

Answer: B

Solution:

Solution:

Let m gram mass of CaCO_3 is required

$$\text{Pure } \text{CaCO}_3 \text{ in } m \text{ gram} = \frac{95}{100} \times m$$

$$\text{Moles of } \text{CaCO}_3 = \frac{95}{100} \times \frac{m}{100}$$

$$\text{Moles of } \text{HCl} \text{ required} = 2 \times \text{moles of } \text{CaCO}_3$$

$$= 2 \times \frac{95}{100} \times \frac{m}{100}$$

$$2 \times \frac{95}{100} \times \frac{m}{100} = \frac{50}{1000} \times 0.5$$

$$m = 1.315\text{g} \approx 1.32\text{g}$$

Question12

An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula

of this compound is : [Atomic wt. of C is 12, H is 1]
[NEET 2021]

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Options:

A. CH

B. CH₂

C. CH₃

D. CH₄

Answer: C

Solution:

Solution:

Element	Mass percentage	No . of mole	Mole ratio
C	78%	$\frac{78}{12} = 6.5$	$\frac{6.5}{6.5} = 1$
H	22%	$\frac{22}{1} = 22$	$\frac{22}{6.5} = 3.38 \approx 3$

Based on above calculation, possible empirical formula is CH₃ .

Question13

Which one of the followings has maximum number of atoms ?
[2020]

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Options:

A. 1 g of Mg(s) [Atomic mass of Mg = 24]

B. 1 g of O₂ [Atomic mass of O = 16]

C. 1 g of Li(s) [Atomic mass of Li = 7]

D. 1 g of Ag(s) [Atomic mass of Ag = 108]

Answer: C

Solution:

Solution:

$$\text{Number of atoms} = \frac{W}{\text{Molar mass}} \times N_A \times \text{atomicity}$$

$$(a) \text{ Number of Mg atoms} = \frac{1}{24} \times N_A \times 1$$

$$(b) \text{ Number of O atoms} = \frac{1}{32} \times N_A \times 2$$

$$(c) \text{ Number of Li atoms} = \frac{1}{7} \times N_A \times 1$$

$$(d) \text{ Number of Ag atoms} = \frac{1}{108} \times N_A \times 1$$

Question14

The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is (NEET 2019)

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Options:

A. 40

B. 10

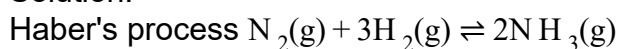
C. 20

D. 30

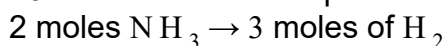
Answer: D

Solution:

Solution:



20 moles needs to be produced



$$\text{Hence } 20 \text{ moles of } NH_3 \rightarrow \frac{3 \times 20}{2} = 30 \text{ of moles of } H_2$$

Question15

The density of 2M aqueous solution of NaOH is 1.28g/cm^3 . The molality of the solution is [Given that molecular mass of NaOH = 40gmol^{-1}]
(Odisha NEET 2019)

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Options:

A. 1.20m

B. 1.56m

C. 1.67m

D. 1.32m

Answer: C

Solution:

Solution:

Density = 1.28g/cc

Conc. of solution = 2M

Molar mass of NaOH = 40gmol^{-1}

Volume of solution = 1L = 1000mL

Mass of solution = $d \times V = 1280\text{g}$

Mass of solute = $n \times \text{Molar mass} = 2 \times 40 = 80\text{g}$

Mass of solvent = $(1280 - 80)\text{g} = 1200\text{g}$

Number of moles of solute = $\frac{80}{40} = 2$

$\therefore \text{Molality} = \frac{2 \times 1000}{1200} = 1.67\text{m}$

Question16

A mixture of 2.3g formic acid and 4.5g oxalic acid is treated with conc. H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be
(NEET 2018)

Options:

A. 1.4

B. 3.0

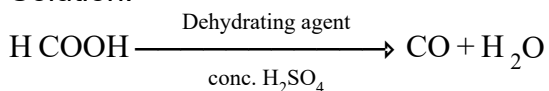
C. 2.8

D. 4.4

Answer: C

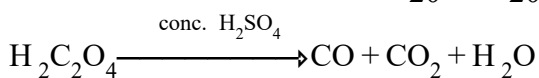
Solution:

Solution:



$$n_i = \frac{2.3}{46} = \frac{1}{20} \quad \quad \quad 0 \quad \quad 0$$

$$n_f = 0 \quad \quad \quad \frac{1}{20} \quad \quad \frac{1}{20}$$



$$n_i = \frac{4.5}{90} = \frac{1}{20} \quad \quad \quad 0 \quad \quad 0 \quad \quad 0$$

$$n_f = 0 \quad \quad \quad \frac{1}{20} \quad \quad \frac{1}{20} \quad \quad \frac{1}{20}$$

H_2O absorbed by H_2SO_4 . Gaseous mixture (containing CO and CO_2) when passed through KOH pellets, CO_2 gets absorbed.

$$\text{Moles of CO left (unabsorbed)} = \frac{1}{20} + \frac{1}{20} = \frac{1}{10}$$

$$\text{Mass of CO} = \text{moles} \times \text{molar mass} = \frac{1}{10} \times 28 = 2.8\text{g}$$

Question17

**In which case is number of molecules of water maximum?
(NEET 2018)**

Options:

A. 18mL of water

B. 0.18 g of water

C. 0.00224L of water vapours at 1 atm and 273K

D. 10^{-3} mol of water

Answer: A

Solution:

Solution:

(a) Mass of water = $V \times d = 18 \times 1 = 18\text{g}$

Molecules of water = mole $\times N_A = \frac{18}{18} N_A = N_A$

(b) Molecules of water = mole $\times N_A = \frac{0.18}{18} N_A$
 $= 10^{-2} N_A$

(c) Moles of water = $\frac{0.00224}{22.4} = 10^{-4}$

Molecules of water = mole $\times N_A = 10^{-4} N_A$

(d) Molecules of water = mole $\times N_A = 10^{-3} N_A$

Question18

Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2 . When 0.1 mole of XY_2 weighs 10g and 0.05mol e of X_3Y_2 weighs 9g, the atomic weights of X and Y are (NEET-II 2016)

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Options:

A. 40,30

B. 60,40

C. 20,30

D. 30,20

Answer: A

Solution:

Solution:

Let atomic weight of element X is x and that of element Y is y.

$$\text{For } XY_2, n = \frac{w}{\text{Mol. wt.}}$$

$$0.1 = \frac{10}{x+2y} \Rightarrow x+2y = \frac{10}{0.1} = 100 \dots (i)$$

$$\text{For } X_3Y_2, n = \frac{w}{\text{Mol. wt.}}$$

$$0.05 = \frac{9}{3x+2y} \Rightarrow 3x+2y = \frac{9}{0.05} = 180 \dots (ii)$$

On solving equations (i) and (ii), we get $y = 30$

$$x + 2(30) = 100 \Rightarrow x = 100 - 60 = 40$$

Question19

The number of water molecules is maximum in (2015)

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Options:

- A. 1.8 gram of water
- B. 18 gram of water
- C. 18 moles of water
- D. 18 molecules of water

Answer: C

Solution:

Solution:

$$1.8 \text{ gram of water} = \frac{6.023 \times 10^{23}}{18} \times 1.8$$

$$= 6.023 \times 10^{22} \text{ molecules}$$

$$18 \text{ grams of water} = 6.023 \times 10^{23} \text{ molecules}$$

$$18 \text{ moles of water} = 18 \times 6.023 \times 10^{23} \text{ molecules}$$

Question20

If Avogadro number N_A , is changed from $6.022 \times 10^{23} \text{ mol}^{-1}$ to $6.022 \times 10^{20} \text{ mol}^{-1}$, this would change
(2015)

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Options:

- A. the mass of one mole of carbon
- B. the ratio of chemical species to each other in a balanced equation
- C. the ratio of elements to each other in a compound
- D. the definition of mass in units of grams

Answer: A

Solution:

Solution:

Mass of 1 mol (6.022×10^{20} atoms) of carbon = 12g

If Avogadro number is changed to 6.022×10^{20} atoms then

$$\text{mass of 1 mol of carbon} = \frac{12 \times 6.022 \times 10^{20}}{6.022 \times 10^{23}} = 12 \times 10^{-3} \text{ g}$$

Question21

What is the mass of the precipitate formed when 50 mL. of 16.9 % solution of AgNO_3 is mixed with 50 mL of 5.8% NaCl solution ?
(Ag = 107.8, N = 14, O = 16, Na = 23 , Cl = 35.5)
(2015)

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Options:

- A. 3.5 g
- B. 7 g
- C. 14 g

D. 28 g

Answer: B

Solution:

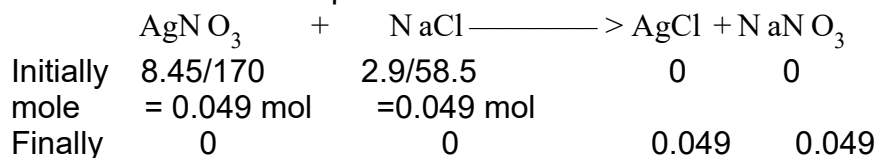
Solution:

16.9% solution of AgNO_3 means 16.9g of AgNO_3 in 100 mL of solution.

16.9 g of AgNO_3 in 100 mL solution = 8.45 g of AgNO_3 in 50 mL solution.

Similarly, 5.8% of NaCl in 100 mL solution = 2.9 g of NaCl in 50 mL solution.

The reaction can be represented as:



$\therefore$ Mass of AgCl precipitated = 0.049×143.3

$-7.02 \approx 7\text{g}$

Question22

A mixtures of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w),What is the ratio of the two gases in the mixture ?
(2015 Cancelled)

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Options:

A. 16 : 1

B. 2 : 1

C. 1 : 4

D. 4 : 1

Answer: D

Solution:

Solution:

Number of moles $\text{H}_2 = \frac{1}{2}$

Number of moles of $\text{O}_2 = \frac{4}{32}$

Hence, molar ratio = $\frac{1}{2} : \frac{4}{32} = 4 : 1$

Question23

1.0 g of magnesium is burnt with 0.56 g O₂ in a closed vessel. Which reactant is left in excess and how much ?

(At. wt. Mg = 24, O = 16)

(2014)

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Options:

A. Mg, 0.16 g

B. O₂, 0.16 g

C. Mg, 0.44 g

D. O₂, 0.28 g

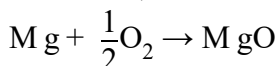
Answer: A

Solution:

Solution:

Given Data, 1.0g of magnesium is burnt with 0.56gO₂

We know, the balanced chemical reaction of the above phenomena is as follows:



We calculate the number of moles of each of the reactants is required, it is given by the formula

$$\text{Number of moles of a substance} = \frac{\text{Mass given}}{\text{Relative formula mass}}.$$

The formula mass of Mg and O₂ are 24 and 32 moles respectively as per the chemical properties of magnesium and oxygen. Hence, the number of moles of Mg and O₂ are $\frac{1.0}{24}$ and $\frac{0.56}{32}$, i.e. $\frac{0.5}{12}$ and $\frac{0.07}{4}$ respectively.

Let us assume 'x' moles of Mg is used up to form MgO, i.e. from the balanced equation, x moles of Magnesium is reacted with $\frac{x}{2}$ moles of Oxygen to form x moles of Magnesium Oxide.

Hence after the reaction is over, the remaining amount of Magnesium Mg is $\frac{0.5}{12} - x$ moles and the

remaining amount of oxygen is $\frac{0.07}{4} - \frac{x}{2}$ moles.

But in this chemical reaction, oxygen is the limiting reagent, i.e. the reaction goes on until oxygen is available. The reaction only ends after all the oxygen available is over. Hence the number of moles of left over oxygen must be equal to zero, $\frac{0.07}{4} - \frac{x}{2} = 0 \Rightarrow x = \frac{0.07}{2}$

Therefore the number of moles of magnesium left over is $\frac{0.5}{12} - x \Rightarrow \frac{0.5}{12} - \frac{0.07}{2} = \frac{1 - 0.07 \times 12}{24}$

moles $\Rightarrow \frac{0.16}{24}$ moles

Hence the mass of magnesium leftover = 0.16g

Thus, when 1.0g of magnesium is burnt with 0.56gO₂ in a closed vessel, 0.16g magnesium is left in excess.

Question24

When 22.4 liters of H₂(g) is mixed with 11.2 liters of Cl₂(g),each at S.T.P, the moles of H Cl (g) formed is equal to (2014)

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Options:

A. 1 mol of H Cl_(g)

B. 2 mol of H Cl_(g)

C. 0.5 mol of H Cl_(g)

D. 1.5 mol of H Cl_(g)

Answer: A

Solution:

Solution:

1 mole = 22.4 litres at S.T.P.

${}^n\text{H}_2 = \frac{22.4}{22.4} = 1\text{mol}$, ${}^n\text{Cl}_2 = \frac{11.2}{22.4} = 0.5\text{mol}$

Reaction is as

	H _{2(g)}	+	Cl _{2(g)}	—————>	2H Cl _(g)
Initial	1mol		0.5mol		0
Final	(1 – 0.5)		(0.5 – 0.5)		2 × 0.5
	= 0.5 mol		= 0 mol		1 mol

Here,Cl₂ is limiting reagent.So,1 mole of H Cl (g) is formed

Question25

Equal masses of H_2 , O_2 and methane have been taken in a container of volume V at temperature $27^\circ C$ in identical conditions. The ratio of the volumes of gases $H_2 : O_2 : \text{methane}$ would be
(2014)

Options:

A. $8 : 16 : 1$

B. $16 : 8 : 1$

C. $16 : 1 : 2$

D. $8 : 1 : 2$

Answer: C

Solution:

Solution:

According to Avogadro's hypothesis, ratio of the volumes of gases will be equal to the ratio of their no. of moles.

So, no. of moles $n = \frac{\text{Mass}}{\text{Mol. mass}}$

$$n_{H_2} = \frac{w}{2}; n_{O_2} = \frac{w}{32}; n_{CH_4} = \frac{w}{16}$$

So, the ratio is $\frac{w}{2} : \frac{w}{32} : \frac{w}{16}$ or $16 : 1 : 2$

Question26

6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is
(2013 NEET)

Options:

- A. 0.001 M
- B. 0.1 M
- C. 0.02 M
- D. 0.01 M

Answer: D

Solution:

Solution:

Molarity (M) = (Number of solute)/(volume of solution in litres)

where in

It is defined as the number of moles of the solute in 1 litre of the solution.

No. of molecules (given) = 6.02×10^{20}

$$\text{No. of moles} = \frac{6.02 \times 10^{20}}{6.02 \times 10^{23}} = 10^{-3} \text{ mol}$$

Volume of solution = 100 ml = 0.1L

Therefore, Molarity = $\frac{\text{no. of moles}}{\text{volume}}$

$$\frac{10^{-3}}{0.1} = 0.01\text{M}$$

Question27

**In an experiment it showed that 10mL of 0.05M solution of chloride required 10mL of 0.1M solution of AgNO_3 , which of the following will be the formula of the chloride (X stands for the symbol of the element other than chlorine)?
(Karnataka NEET 2013)**

Options:

- A. X_2Cl_2
- B. XCl_2
- C. XCl_4

D. $X_2 Cl$

Answer: B

Solution:

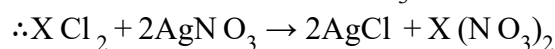
Solution:

Millimoles of solution of chloride

$$= 0.05 \times 10 = 0.5$$

Millimoles of $AgNO_3$ solution = $10 \times 0.1 = 1$

So, the millimoles of $AgNO_3$ are double than the chloride solution.



Question28

Which has the maximum number of molecules among the following ?

(2011 Mains)

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Options:

A. 44 g CO_2

B. 48 g O_3

C. 8 g H_2

D. 64 g SO_2

Answer: C

Solution:

Solution:

$8gH_2$ has 4 moles while the others has 1 mole each

Question29

25.3 g of sodium carbonate Na_2CO_3 is dissolved in enough water to make 250 mL of solution. If sodium carbonate dissociates completely, molar concentration of sodium ion, Na^+ and carbonate ions, CO_3^{2-} are respectively

**(Molar mass of $\text{Na}_2\text{CO}_3 = 106 \text{ g mol}^{-1}$)
(2010)**

©

Options:

- A. 0.955 M and 1.910 M
- B. 1.910 M and 0.955 M
- C. 1.90 M and 1.910 M
- D. 0.477 and 0.477 M

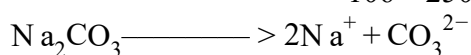
Answer: B

Solution:

Solution:

Given that molar mass of $\text{Na}_2\text{CO}_3 = 106\text{g}$

$$\therefore \text{Molarity of solution} = \frac{2.53 \times 1000}{106 \times 250} = 0.955\text{M}$$



$$[\text{Na}^+] = 2[\text{Na}_2\text{CO}_3] = 2 \times 0.955 = 1.910\text{M}$$

$$[\text{CO}_3^{2-}] = [\text{Na}_2\text{CO}_3] = 0.955\text{M}$$

Question 30

The number of atoms in 0.1 mol of a triatomic gas is

$$N_A \times 6.02 \times 10^{23} \text{ mol}^{-1}$$

(2010)

©

Options:

A. 6.026×10^{22}

B. 1.806×10^{23}

C. 3.600×10^{21}

D. 1.800×10^{22}

Answer: B

Solution:

Solution:

$$\text{No. of atoms} = N_A \times \text{No. of moles} \times 3$$

$$-6.023 \times 10^{23} \times 0.1 \times 3 - 1.806 \times 10^{23}$$

Question31

10 g of hydrogen and 64 g of oxygen were filled in a steel vessel and exploded. Amount of water produced in this reaction will be (2009)

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Options:

A. 3 mol

B. 4 mol

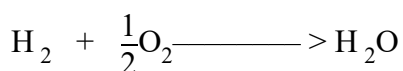
C. 1 mol

D. 2 mol

Answer: B

Solution:

Solution:



2g	16g	18 g
1 mol	0.5 mol	1 mol

10 g of H_2 – 5 mol and 64 g of O_2 = 2mol

∴ In this reaction, oxygen is the limiting reagent so amount of H_2O produced depends on that of O_2
Since 0.5 mol of O_2 gives 1 mol H_2O
∴ 2 mol of O_2 will give 4 mol H_2O

Question32

An organic compound contains carbon, hydrogen and oxygen, Its elemental analysis gave C, 38.71% and H, 9.67%. The empirical formula of the compound would be (2008)

Options:

- A. CH O
- B. CH_4O
- C. CH_3O
- D. CH_2O

Answer: C

Solution:

Solution:

Element	%	Atomic mass	Mole ratio	Simple ratio
C	38.71	12	$\frac{38.71}{12} = 3.22$	$\frac{3.22}{3.22} = 1$
H	9.67	1	$\frac{9.67}{1} = 9.67$	$\frac{9.67}{3.22} = 3$
O	51.62	16	$\frac{51.62}{16} = 3.22$	$\frac{3.22}{3.22} = 1$

Hence empirical formula of the compound would be CH_3O

Question33

**How many moles of lead (II) chloride will be formed from a reaction between 6.5 g of PbO and 3.2 g HCl ?
(2008)**

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Options:

A. 0.011

B. 0.029

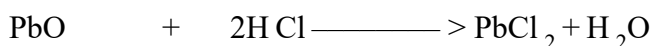
C. 0.044

D. 0.333

Answer: B

Solution:

Solution:



$$\frac{6.5}{224} \text{ mol} \quad \frac{3.2}{36.5} \text{ mol}$$
$$= 0.029 \text{ mol} \quad = 0.087 \text{ mol}$$

Formation of moles of lead (II) chloride depends upon the no of moles of PbO which acts as a limiting factor here, So no. of moles of PbCl_2 formed will be equal to the no of moles of PbO i.e., 0.029

Question34

**What volume of oxygen gas (O_2) measured at 0°C and 1 atm, is needed to burn completely 1 L propane gas (C_3H_8) measured under the same conditions ?
(2008)**

©

Options:

A. 5 L

B. 10 L

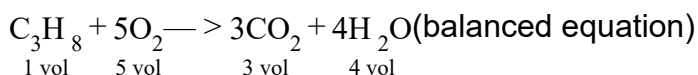
C. 7 L

D. 6 L

Answer: A

Solution:

Solution:



According to the above equation

1 vol. or 1 litre of propane requires to 5 vol. or 5 litre of O₂ to burn completely.

Question35

An element ,X the following isotopic composition :

²⁰⁰X : 90% ¹⁹⁹X : 8.0% ²⁰²X ; 2.0%

**The weighted average atomic mass of the naturally occurring element X is closest to
(2007)**

Options:

A. 201 amu

B. 202 amu

C. 199 amu

D. 200 amu

Answer: D

Solution:

Solution:

Average isotopic mass of X

$$= \frac{200 \times 90 + 199 \times 8 + 202 \times 2}{90 + 8 + 2}$$

$$= \frac{18000 + 1592 + 404}{100} = 199.96 \text{ a . m . u} \approx 200 \text{ a . m . u}$$

Question36

The maximum number of molecules is present in
(2004)

Options:

A. 15L of H_2 gas at STP

B. 5L of N_2 gas at STP

C. 0.5g of H_2 gas

D. 10g of O_2 gas.

Answer: A

Solution:

Solution:

At ST P, $22.4L = 6.023 \times 10^{23}$ molecules

$$15LH_2 = \frac{6.023 \times 10^{23} \times 15}{22.4} = 4.033 \times 10^{23}$$

$$5LN_2 = \frac{6.023 \times 10^{23} \times 5}{22.4} = 1.344 \times 10^{23}$$

$$2gH_2 = 6.023 \times 10^{23}$$

$$0.5gH_2 = \frac{6.023 \times 10^{23} \times 0.5}{2} = 1.505 \times 10^{23}$$

$$32gO_2 = 6.023 \times 10^{23}$$

10g of

$$O_2 = \frac{6.023 \times 10^{23} \times 10}{32} = 1.882 \times 10^{23}$$

Question37

Which has maximum molecules?
(2002)

Options:

A. 7 g N₂

B. 2 g H₂

C. 16 g NO₂

D. 16 g O₂

Answer: B

Solution:

Solution:

Number of molecules = moles $\times$ N_A

Molecules of N₂ = $\frac{7}{14}$ N_A = 0.5 N_A

Molecules of H₂ = 2 N_A

Molecules of NO₂ = $\frac{16}{46}$ N_A = 0.35 N_A

Molecules of O₂ = $\frac{16}{32}$ N_A = 0.5 N_A

$\therefore$ 2g H₂ (1g mole H₂) contains maximum molecules.

Question38

Percentage of Se in peroxidase anhydrous enzyme is 0.5% by weight (at. wt. = 78.4) then minimum molecular weight of peroxidase anhydrous enzyme is (2001)

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Options:

A. 1.568×10^4

B. 1.568×10^3

C. 15.68

D. 2.136×10^4

Answer: A

Solution:

Solution:

In peroxidase anhydrous enzyme 0.5% Se is present means, 0.5g Se is present in 100g of enzyme. In a molecule of enzyme one Se atom must be present. Hence, 78.4g Se will be present in

$$\frac{100}{0.5} \times 78.4 = 1.568 \times 10^4$$

Question39

Molarity of liquid HCl, if density of solution is 1.17g/cc is (2001)

©

Options:

- A. 36.5
- B. 18.25
- C. 32.05
- D. 42.10

Answer: C

Solution:

Solution:

Density = 1.17g/cc

⇒ 1cc. solution contains 1.17g of HCl

$$\therefore \text{Molarity} = \frac{1.17 \times 1000}{36.5 \times 1} = 32.05$$

Question40

Specific volume of cylindrical virus particle is 6.02×10^{-2} cc/g whose radius and length are 7Å and 10Å respectively. If $N_A = 6.02 \times 10^{23}$ find molecular weight of virus.

(2001)

©

Options:

A. 15.4kg/mol

B. 1.54×10^4 kg/mol

C. 3.08×10^4 kg/mol

D. 3.08×10^3 kg/mol

Answer: A

Solution:

Solution:

Specific volume (vol. of 1g) of cylindrical virus particle = 6.02×10^{-2} cc/g

Radius of virus, $r = 7\text{\AA} = 7 \times 10^{-8}$ cm

Volume of virus = $\pi r^2 l$

$$= \frac{22}{7} \times (7 \times 10^{-8})^2 \times 10 \times 10^{-8} = 154 \times 10^{-23} \text{cc}$$

$$\text{wt. of one virus particle} = \frac{\text{Volume}}{\text{Specific volume}}$$

$$= \frac{154 \times 10^{-23}}{6.02 \times 10^{-2}} \text{g}$$

$\therefore$ Molecular wt. of virus = wt. of N_A particles

$$= \frac{154 \times 10^{-23}}{6.02 \times 10^{-2}} \times 6.02 \times 10^{-23} \text{g/mol}$$

$$= 15400 \text{g/mol} = 15.4 \text{kg/mol}$$

Question41

Volume of CO_2 obtained by the complete decomposition of 9.85g of BaCO_3 is

(2000)

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Options:

A. 2.24L

B. 1.12L

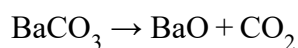
C. 0.84L

D. 0.56L

Answer: B

Solution:

Solution:



197.34 g 22.4 L at N.T.P

$$9.85\text{g} \quad \frac{22.4}{197.34} \times 9.85$$

$$= 1.118 \text{ L}$$

$\Rightarrow 9.85\text{g BaCO}_3$ will produce 1.118 L CO_2 at N.T.P. on the complete decomposition.

Question42

Oxidation numbers of A, B, C are +2,+5 and -2 respectively. Possible formula of compound is (2000)

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Options:

A. $\text{A}_2(\text{BC}_2)_2$

B. $\text{A}_3(\text{BC}_4)_2$

C. $\text{A}_2(\text{BC}_3)_2$

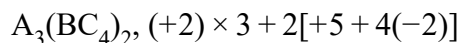
D. $\text{A}_3(\text{B}_2\text{C})_2$

Answer: B

Solution:

Solution:

In



$$\Rightarrow +6 + 2(-3) = 0$$

Hence, in the compound $A_3(BC_4)_2$, the oxidation no. of 'A', 'B' and 'C' are +2, +5 and -2 respectively.

Question43

The number of atoms in 4.25g of NH_3 is approximately
(1999)

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Options:

A. 4×10^{23}

B. 2×10^{23}

C. 1×10^{23}

D. 6×10^{23}

Answer: D

Solution:

Solution:

No. of molecules in 4.25g NH_3

$$= \frac{4.25}{17} \times 6.023 \times 10^{23} = 2.5 \times 6.023 \times 10^{22}$$

Number of atoms in 4.25g NH_3

$$= 4 \times 2.5 \times 6.023 \times 10^{22} = 6.023 \times 10^{23}$$

Question44

Given the numbers : 161cm, 0.161cm, 0.0161cm. The number of significant figures for the three numbers is
(1998)

©

Options:

- A. 3, 3 and 4 respectively
- B. 3, 4 and 4 respectively
- C. 3, 4 and 5 respectively
- D. 3, 3 and 3 respectively.

Answer: D

Solution:

Solution:

Zeros placed left to the number are never significant, therefore the no. of significant figures for the numbers.

161cm = 0.161cm and 0.0161cm are same, i . e., 3

Question45

Haemoglobin contains 0.334% of iron by weight. The molecular weight of haemoglobin is approximately 67200. The number of iron atoms (Atomic weight of Fe is 56) present in one molecule of haemoglobin is (1998)

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Options:

- A. 4
- B. 6
- C. 3
- D. 2

Answer: A

Solution:

Solution:

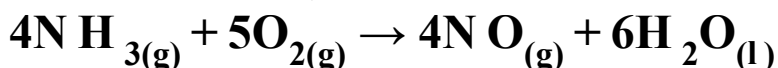
Quantity of iron in one molecule

$$= \frac{67200}{100} \times 0.334 = 224.45 \text{amu}$$

$$\text{No. of iron atoms in one molecule of haemoglobin} = \frac{224.45}{56} = 4$$

Question46

In the reaction,



when 1 mole of ammonia and 1 mole of O_2 are made to react to completion :
(1998)

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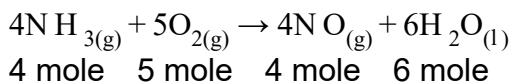
Options:

- A. all the oxygen will be consumed
- B. 1.0 mole of NO will be produced
- C. 1.0 mole of H_2O is produced
- D. all the ammonia will be consumed.

Answer: A

Solution:

Solution:



$$\Rightarrow 1 \text{ mole of N H}_3 \text{ requires } = \frac{5}{4} = 1.25 \text{ mole of}$$

$$\text{oxygen while 1 mole of O}_2 \text{ requires } = \frac{4}{5} = 0.8 \text{ mole of N H}_3$$

Therefore, all oxygen will be consumed.

Question47

0.24g of a volatile gas, upon vaporisation, gives 45mL vapour at NTP. What will be the vapour density of the substance? (Density of $H_2 = 0.089\text{g/L}$)
(1996)

©

Options:

- A. 95.93
- B. 59.93
- C. 95.39
- D. 5.993

Answer: B

Solution:

Solution:

Weight of gas = 0.24g

Volume of gas = 45mL = 0.045 litre and density of $H_2 = 0.089$

weight of 45mL of H_2 = density $\times$ volume = $0.089 \times 0.045 = 4.005 \times 10^{-3}\text{g}$

Therefore, vapour density

$$\begin{aligned} &= \frac{\text{Weight of certain volume of substance}}{\text{Weight of same volume of hydrogen}} \\ &= \frac{0.24}{4.005 \times 10^{-3}} = 59.93 \end{aligned}$$

Question48

The amount of zinc required to produce 224mL of H_2 at ST P on treatment with dilute H_2SO_4 will be
(1996)

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Options:

- A. 65g

B. 0.065g

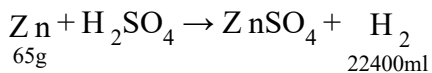
C. 0.65g

D. 6.5g

Answer: C

Solution:

Solution:



since 65g of zinc reacts to liberate 22400mL of H₂ at ST P, therefore amount of zinc needed to produce 224mL of H₂ at STP

$$= \frac{65}{22400} \times 224 = 0.65\text{g}$$

Question49

The dimensions of pressure are the same as that of (1995)

©

Options:

A. force per unit volume

B. energy per unit volume

C. force

D. energy.

Answer: B

Solution:

Solution:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Therefore, dimensions of pressure} = \frac{\text{MLT}^{-2}}{\text{L}^2}$$

$$= \text{ML}^{-1}\text{T}^{-2}$$

and dimensions of energy per unit volume

$$= \frac{\text{Energy}}{\text{Volume}} = \frac{\text{ML}^2\text{T}^{-2}}{\text{L}^3} = \text{ML}^{-1}\text{T}^{-2}$$

Question50

The number of moles of oxygen in one litre of air containing 21% oxygen by volume, under standard conditions, is (1995)

©

Options:

A. 0.0093mol

B. 2.10mol

C. 0.186mol

D. 0.21 mol.

Answer: A

Solution:

Solution:

Volume of oxygen in one litre of air

$$= \frac{21}{100} \times 1000 = 210\text{mL}$$

$$\text{Therefore, no. of mol} = \frac{210}{22400} = 0.0093\text{mol}$$

Question51

The total number of valence electrons in 4.2g of N_3^- ion is (N_A is the Avogadro's number) (1994)

©

Options:

A. $2.1N_A$

B. $4.2N_A$

C. $1.6N_A$

D. $3.2N_A$

Answer: C

Solution:

Solution:

Each nitrogen atom has 5 valence electrons, therefore total number of electrons in N_3^- ion is 16.

since the molecular mass of N_3 is 42, therefore total number of electrons in 4.2g of N_3^- ion

$$= \frac{4.2}{42} \times 16 \times N_A = 1.6N_A$$

Question52

A 5 molar solution of H_2SO_4 is diluted from 1 litre to a volume of 10 litres, the normality of the solution will be (1991)

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Options:

A. 1N

B. 0.1N

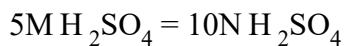
C. 5N

D. 0.5N

Answer: A

Solution:

Solution:



$$N_1 V_1 = N_2 V_2 \Rightarrow 10 \times 1 = N_2 \times 10 \Rightarrow N_2 = 1N$$

Question53

**The number of gram molecules of oxygen in 6.02×10^{24} CO molecules is
(1990)**

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Options:

- A. 10g molecules
- B. 5g molecules
- C. 1g molecules
- D. 0.5g molecules.

Answer: B

Solution:

Solution:

Avogadro's No., $N_A = 6.02 \times 10^{23}$ molecules

$$\begin{aligned} \therefore 6.02 \times 10^{24} \text{CO molecules} &= 10 \text{ moles CO} \\ &= 10 \text{g atoms of O} = 5 \text{g molecules of O}_2 \end{aligned}$$

Question54

**Boron has two stable isotopes, ^{10}B (19%) and ^{11}B (81%). Calculate average at. wt. of boron in the periodic table.
(1990)**

©

Options:

- A. 10.8
- B. 10.2
- C. 11.2
- D. 10.0

Answer: A

Solution:

Solution:

$$\text{Average atomic mass} = \frac{19 \times 10 + 81 \times 11}{100} = 10.81$$

Question 55

The molecular weight of O_2 and SO_2 are 32 and 64 respectively. At 15°C and 150 mmHg pressure, one litre of O_2 contains "N" molecules. The number of molecules in two litres of SO_2 under the same conditions of temperature and pressure will be (1990)

©

Options:

- A. $\frac{N}{2}$
- B. N
- C. 2N
- D. 4N

Answer: C

Solution:

Solution:

If 1L of one gas contains N molecules, 2L of any gas under the same conditions will contain 2N molecules.

Question56

A metal oxide has the formula Z_2O_3 . It can be reduced by hydrogen to give free metal and water. 0.1596g of the metal oxide requires 6mg of hydrogen for complete reduction. The atomic weight of the metal is (1989)

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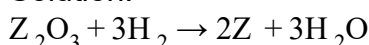
Options:

- A. 27.9
- B. 159.6
- C. 79.8
- D. 55.8

Answer: D

Solution:

Solution:



Valency of metal in $Z_2O_3 = 3$

0.1596g of Z_2O_3 react with 6mg of H_2 .

$$[1\text{mg} = 0.001\text{g} = 10^{-3}\text{g}]$$

$$\therefore 1\text{g of } H_2 \text{ react with } = \frac{0.1596}{0.006} = 26.6\text{g of } Z_2O_3$$

$$\therefore \text{Eq. wt. of } Z_2O_3 = 26.6$$

$$\text{Now, Eq. wt. of } Z + \text{Eq. wt. of } O = \text{Eq. wt. of } Z_2O_3 = 26.6$$

$$\Rightarrow \text{Eq. wt. of } Z = 26.6 - 8 = 18.6$$

$$\therefore \text{At. wt. of } Z = 18.6 \times 3 = 55.8$$

$$\left[\because \text{Eq. wt.} = \frac{\text{Atomic wt.}}{\text{Valency of metal}} \right]$$

Question57

Ratio of C_p and C_v of a gas X is 1.4. The number of atoms of the gas X " present in 11.2 litres of it at NTP will be (1989)

©

Options:

A. 6.02×10^{23}

B. 1.2×10^{23}

C. 3.01×10^{23}

D. 2.01×10^{23}

Answer: A

Solution:

Solution:

Here, $C_p/C_v = 1.4$, which shows that the gas is diatomic

$22.4\text{L at NTP} = 6.02 \times 10^{23} \text{ molecules}$

$\therefore 11.2\text{L at NTP} = 3.01 \times 10^{23} \text{ molecules}$

since gas is diatomic.

$\therefore 11.2\text{L at NTP} = 6.02 \times 10^{23} \text{ atom}$

Question58

What is the weight of oxygen required for the complete combustion of 2.8kg of ethylene? (1989)

©

Options:

A. 2.8kg

B. 6.4kg

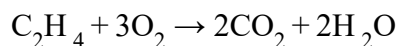
C. 9.6kg

D. 96kg

Answer: C

Solution:

Solution:



28g 96g

For complete combustion

2.8kg of C_2H_4 requires

$$= \frac{96\text{g}}{28\text{g}} \times 2.8\text{kg of O}_2 = \frac{96}{28} \times 2.8 \times 10^3\text{g}$$

$$= 9.6 \times 10^3\text{g} = 9.6\text{kg of O}_2$$

Question59

**The number of oxygen atoms in 4.4g of CO_2 is
(1989)**

©

Options:

A. 1.2×10^{23}

B. 6×10^{22}

C. 6×10^{23}

D. 12×10^{23}

Answer: A

Solution:

Solution:

$$1 \text{ mol of CO}_2 = 44\text{g of CO}_2$$

$$\therefore 4.4\text{gCO}_2 = 0.1 \text{ mol CO}_2 = 6 \times 10^{22} \text{ molecules}$$

$$[\text{since, 1 mole CO}_2 = 6 \times 10^{23} \text{ molecules}]$$

$$= 2 \times 6 \times 10^{22} \text{ atoms of O} = 1.2 \times 10^{23} \text{ atoms of O}$$

Question60

At S.T.P. the density of CCl_4 vapour in g/L will be nearest to (1988)

Options:

- A. 6.87
- B. 3.42
- C. 10.26
- D. 4.57

Answer: A

Solution:

Solution:

Weight of 1 mol CCl_4 vapour

$$= 12 + 4 \times 35.5 = 154\text{g}$$

$$\therefore \text{Density of } \text{CCl}_4 \text{ vapour} = \frac{154}{22.4} \text{gL}^{-1}$$

$$= 6.875 \text{gL}^{-1}$$

Question61

One litre hard water contains 12.00mg M g^{2+} . Milli-equivalents of washing soda required to remove its hardness is (1988)

Options:

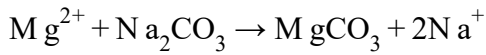
- A. 1
- B. 12.16
- C. 1×10^{-3}

D. 12.16×10^{-3}

Answer: A

Solution:

Solution:



1g eq. 1geq.

1g eq. of $\text{Mg}^{2+} = 12\text{g}$ of $\text{Mg}^{2+} = 12000\text{mg}$

Now, 1000 millieq. of $\text{Na}_2\text{CO}_3 = 12000\text{mg}$ of Mg^{2+}

$\therefore$ 1 millieq. of $\text{Na}_2\text{CO}_3 = 12\text{mg}$ of Mg^{2+}

Question62

**1cc N_2O at N T P contains
(1988)**

©

Options:

A. $\frac{1.8}{224} \times 10^{22}$ atoms

B. $\frac{6.02}{22400} \times 10^{23}$ molecules

C. $\frac{1.32}{224} \times 10^{23}$ electrons

D. all of the above.

Answer: D

Solution:

Solution:

As we know

22400cc of N_2O contain 6.02×10^{23} molecules

$\therefore$ 1 cc of N_2O contain $\frac{6.02 \times 10^{23}}{22400}$ molecules

since in N_2O molecule there are 3 atoms

$$\therefore 1 \text{ cc N}_2\text{O} = \frac{3 \times 6.02 \times 10^{23}}{22400} \text{ atoms}$$
$$= \frac{1.8 \times 10^{22}}{224} \text{ atoms}$$

No. of electrons in a molecule of

$$\text{N}_2\text{O} = 7 + 7 + 8 = 22$$

$$\text{Hence, no. of electrons} = \frac{6.02 \times 10^{23}}{22400} \times 22 \text{ electron}$$

$$= \frac{1.32}{224} \times 10^{23} \text{ electrons}$$
